DAY 07 ASSIGNMENT

IPv4 Addressing & IP Configuration

Submitted By: SREEKAKULAM THULASIRAM

1) THEORY

1. What is a network address?

A network address refers to an IPv4 address where all host bits have been turned to 0. It is used to refer to the network.

2. What is a broadcast address?

A broadcast address refers to an IPv4 address where all host bits have been turned to 1. Used to broadcast data to all devices in a specific SUBNET.

3. What is a valid host address?

A valid host address refers to an IPv4 address that can be assigned to an individual device within a SUBNET. Can be used by computers, printers, routers and other network devices to communicate. Located between network and broadcast address in a standard SUBNET.

4. What is the first valid IP address?

Normally the first valid IP address will be directly after the network address.

5. What is the last valid IP address?

Normally the last valid IP address will be directly before the broadcast address.

6. What does /24 mean?

/24 refers to the number of bits used for the network part of the IPv4 address. In this case there are 24 bits used for the network and 8 for host.

7. What is the SUBNET MASK for /24?

The SUBNET MASK for /24 is:

255.255.255.0

There are 24 network bits and 8 host bits.

8. What is the default gateway?

A default gateway is typically a router that will route traffic from YOUR local network. When trying to communicate to an address that is not on YOUR local SUBNET, the traffic will go via the default gateway.

9. Why would a router interface require an IP address?

An IP address will be required on a router interface so that it can communicate with devices on the network it is connected to. This will allow it to receive and forward IP packets, and can be used as a default gateway for devices on the network.

10. What is a static IP configuration?

A static IP configuration means that the network settings are manually configured or permanently assigned. This could include IP address, SUBNET MASK, default gateway and DNS.

11. What does 'show IP interface brief' show?

The 'show IP interface brief' command is in Cisco IOS. It gives a quick overview of the networks interfaces on YOUR router or switch. It will show the interface name, IP address assigned, how it has been configured, interface status and line protocol status.

12. What does 'no shutdown' do to a Cisco interface?

The 'no shutdown' command will enable the network interface on a Cisco device. This removes the shutdown status of the interface and will allow it to operate once required network and physical conditions have been met.

13. What does 'up/up' usually mean for interface status?

'up/up' usually means that the interface is physically operating and the line protocol is working. However, it cannot be guaranteed that you will be able to communicate with all devices on the network.

14. Why should network configuration always be checked?

You should always check YOUR network configuration to ensure you have assigned the correct settings such as IP address, SUBNET MASK, default gateway and DNS. It will also help you identify any errors in YOUR configuration, connection issues and problems that might cause devices to have issues communicating.

15. Why is IPv4 configuration important in Cybersecurity?

IPv4 configuration is important in Cybersecurity as it allows you to identify devices and understand how they communicate. Knowing YOUR IP address, SUBNET and gateway will help you monitor YOUR network, identify any unknown devices and spot any suspicious connections. This can help you investigate any potential cyber attacks.

2) IPv4 EXERCISE

Network: 192.168.100.0/24

As the prefix is /24 there are 24 network bits and 8 host bits.

Subnet Mask: 255.255.255.0

Requirement	Calculated Answer
Network Address	192.168.100.0
First Valid IP	192.168.100.1
Last Valid IP	192.168.100.254
Broadcast Address	192.168.100.255

Network Address - All host bits turned to 0.

First Valid IP - Network Address + 1.

Broadcast Address - All host bits turned to 1.

Last Valid IP - Broadcast Address - 1.

There are 256 IPv4 addresses in this /24 range. However, typically there are only 254 valid host addresses.

3) PRACTICAL

I used the Windows CMD and ran the 'ipconfig /all' command to see what the network configuration was on my active network adapter.

Below is the information I found on my computer.

Network Information	My Computer
IPv4 Address	10.68.248.97
Subnet Mask	255.255.255.0
Default Gateway	10.68.248.199
DNS Server	10.68.248.199

4) SCREENSHOTS

1. IP Configuration

This shows the output from the 'ipconfig /all' command. It shows the network configuration of my active network adapter. Including my IPv4 Address, Subnet Mask, Default Gateway and DNS Server.

```
PS C:\Users\Thulasiram> ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : ThulasiramPc
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 3:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
   Physical Address. . . . . . . . . : 48-51-C5-A7-F6-67
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

```
Wireless LAN adapter Wi-Fi 2:

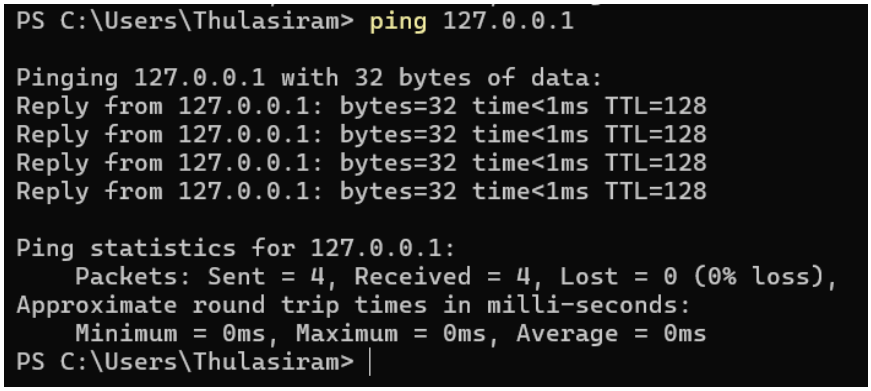
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wireless-AC 9560 160MHz
   Physical Address. . . . . . . . . : 48-51-C5-A7-F6-66
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2401:4900:4d01:f6c9:1661:e786:433e:746f(Pre
   Temporary IPv6 Address. . . . . . : 2401:4900:4d01:f6c9:559e:c651:49cb:eb64(Pre
   Link-local IPv6 Address . . . . . : fe80::6b34:68ed:a853:a8f7%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.68.248.97(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Saturday, 7 November 2026 4:38:52 pm
   Lease Expires . . . . . . . . . . : Saturday, 7 November 2026 7:05:58 pm
   Default Gateway . . . . . . . . . : fe80::cc64:88ff:fe55:a991%15
                                       10.68.248.199
   DHCP Server . . . . . . . . . . . : 10.68.248.199
   DHCPv6 IAID . . . . . . . . . . . : 155734469
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2B-10-43-5A-00-E0-4C-68-0A-A9
   DNS Servers . . . . . . . . . . . : 10.68.248.199
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Hyper-V Virtual Ethernet Adapter
   Physical Address. . . . . . . . . : 00-15-5D-61-18-A5
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::2598:9ba0:9568:cb8%45(Preferred)
   IPv4 Address. . . . . . . . . . . : 172.27.16.1(Preferred)
```

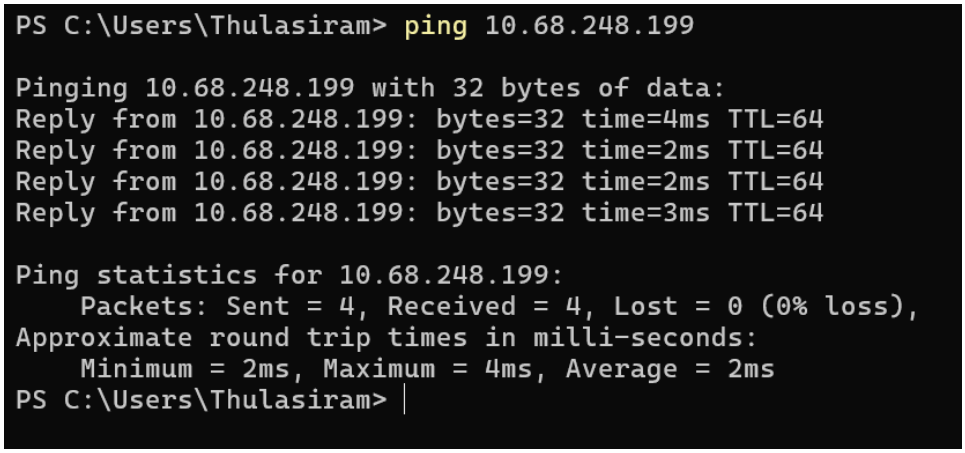
2. Ping Loopback

This shows the output from the 'ping 127.0.0.1' command. It shows I have run the loopback test to see if my local TCP/IP networking stack is functioning.



3. Ping Default Gateway

This shows the output from the ping command using my real default gateway. It helps me see if I can communicate with the local router.



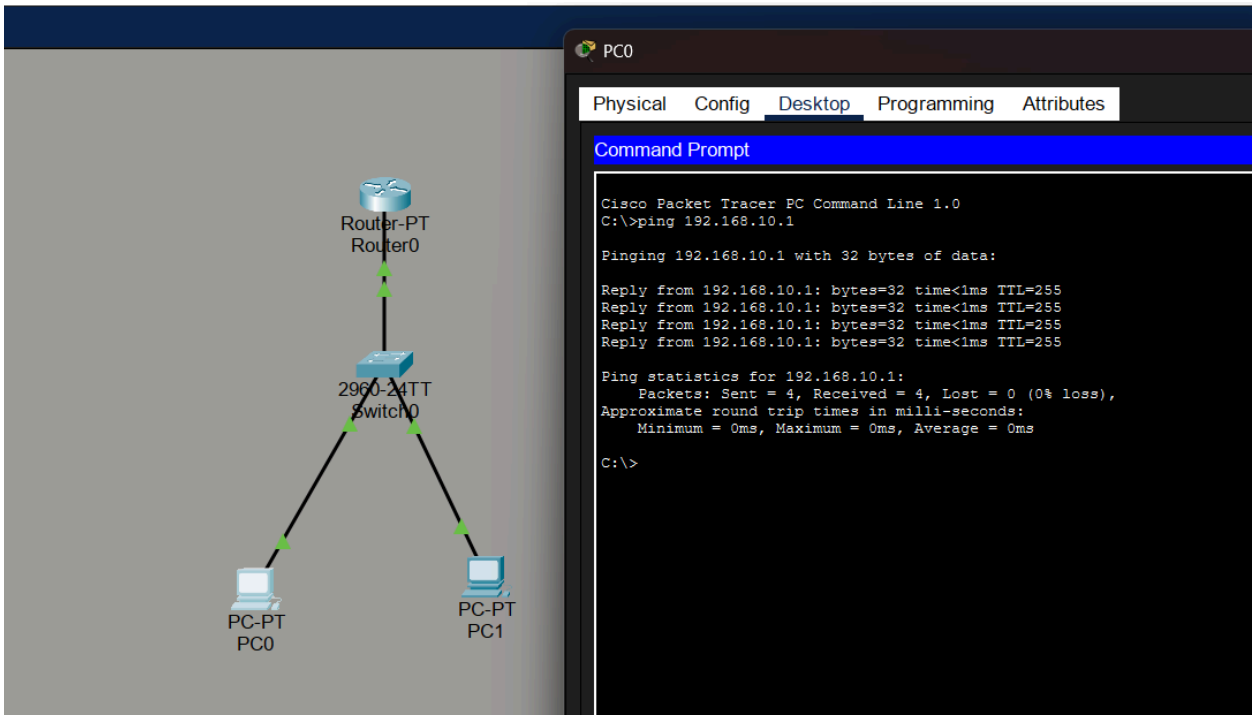
4. Packet Tracer Topology (If completed part 4).

This shows the topology I have created in Cisco Packet Tracer.

```
Router>
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
Router(config)#
```



CONCLUSION

By completing this assignment I have learnt about IPv4 addresses, network addresses, broadcast addresses, valid host addresses and SUBNET masks. I have understood how the /24 prefix works and how to calculate the network address, first valid IP, last valid IP and broadcast address. I have also learnt about default gateways, router interfaces and static IP configurations..